

NAMAN YESHWANTH KUMAR

linkedin.com/in/namany | github.com/naman-ranka | nyeshwan@asu.edu | (623) 249-1265

SUMMARY

Full-stack engineer who sole-builds and ships production AI systems end-to-end (MS Computer Engineering, ASU — GPA 3.81, May 2026). Built and launched Trezzit (app.trezzit.com) — a live SaaS serving **600+ users** for 12+ months, iterating an LLM pipeline from **60% → 95%+** accuracy across 4 providers — and a 21-tool autonomous LangGraph agent with self-correction and multi-provider routing. Owns every layer: React frontend, Django/FastAPI APIs, LLM orchestration, and production infrastructure.

TECHNICAL SKILLS

AI & LLM Systems: LangGraph | LangChain | RAG Pipelines | MCP Protocol | Multi-provider LLM Routing | Prompt Engineering | Tool Use / Function Calling | Pydantic Validation

Full-Stack: React 19 | TypeScript | Next.js | Python | Django REST Framework | FastAPI | PostgreSQL | Redis | Celery | WebSocket

Infrastructure: Docker | DigitalOcean | AWS | Vercel | Supabase | Nginx | GitHub Actions CI/CD | Linux

PROJECT EXPERIENCE

Trezzit — Production AI SaaS Platform (app.trezzit.com)

Feb 2025 – Present

- Solo-built and shipped Trezzit (app.trezzit.com), a production bill-splitting SaaS serving **600+ users** over 12+ months — **341 React components, 60+ REST endpoints, 40 DB models, 995+ backend tests**; Django 5.1 + DRF, React 19 PWA, PostgreSQL (Supabase), Redis, Celery, DigitalOcean VPS (prod) + AWS Lightsail (staging), Vercel; GitHub Actions CI/CD in ~3 min.
- Built two-stage AI receipt scanning pipeline (Gemini, OpenAI, Claude, Grok) that improved parsing accuracy from **~60% → 95%+** — semantic validation with automatic fallback to higher-capability model on failure; pre-computed balance system with O(1) debt reads, incremental updates, and automatic reconciliation.
- Built MCP server (Model Context Protocol) enabling AI assistants to create and query itemized expenses; production monitoring tracks per-provider LLM costs and API response times across all 60+ endpoints.

HEAL.AI — AI Healthcare Billing Assistant

Sep – Nov 2025

- Won Devlabs Hackathon** with HEAL.AI, an AI healthcare billing assistant; built RAG pipeline — PDF chunking and document search over uploaded insurance policies — enabling natural language Q&A, billing overcharge detection, and automated dispute email generation via Gemini 2.5; FastAPI + Next.js + TypeScript, Dockerized.

SiliconCrew — Autonomous LLM Agent for RTL Design

Nov 2025 – Present

- Built SiliconCrew, an autonomous LangGraph agent (21 tools) that takes natural language specs, generates code, self-corrects on test failure via diagnostic analysis, and drives an end-to-end automation pipeline; multi-provider LLM routing (Gemini, OpenAI, Claude) with quality-based fallback; MCP server for VS Code and Claude Desktop integration; FastAPI + WebSocket + Next.js.

Intel Automated Self-Checkout — Open Source Contribution

Jan – Feb 2025

- Contributed sensor analytics microservice to **Intel's official self-checkout reference architecture** — configurable data pipeline (MQTT, Kafka, HTTP) for real-time LiDAR and weight sensor data with Grafana dashboard; **PR #655 merged upstream** after Intel engineering review.

WORK EXPERIENCE

Software Developer (Sole Engineer) — Order Management & Supply Chain

Oct 2022 – Apr 2024

NCR GOLD

Bengaluru, India

- Engineered a production Django platform (PostgreSQL + Celery + Nginx on DigitalOcean) processing **18,000+ orders over 3 years**; built 3 role-based portals (admin, artisan, customer) with multi-method auth — automated PDF invoice generation and WhatsApp Business API delivery on every state transition, eliminating **~20 min** of manual relay per order.
- Reverse-engineered third-party Jewely ERP (**zero public API**) through direct SQL schema analysis; built bidirectional bridge model syncing production weights and ERP status in real time; engineered inbound WhatsApp state-machine chatbot (JSON state per user, admin + customer flows) — full accountability loop across all three portals without disrupting legacy system.

EDUCATION

Master of Science — Computer Engineering (GPA 3.81/4.0)

Expected May 2026

Arizona State University — Coursework: Artificial Intelligence · Foundations of Algorithms · Reinforcement Learning in Robotics · Perception in Robotics · Algorithm/Hardware Co-Design

Tempe, AZ

Bachelor of Engineering — Electrical and Electronics

August 2022

BMS College of Engineering

Bengaluru, India

AWARDS & HONORS

- Won **'Best Use of AI'** at Strategy X DevHacks'25 (ASU) — built AdaptED AI, a Gemini-powered platform that generates personalized flashcards, quizzes, and learning paths from uploaded documents in a 24-hour hackathon.
- Won **1st place at Devlabs Hackathon** — HEAL.AI, an AI healthcare assistant with a custom RAG pipeline for insurance Q&A and automated medical bill dispute generation.
- Secured **2nd place at Hack SoDA 2024** (Amazon-sponsored, 24-hour) — built PassGen, a secure offline Chrome Extension for unique password generation.